

Peripherals: Working Document

Digital Audio Player • Rev 1

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Contents

1	Summary	1
2	Design Requirements	1
3	Interface	1
4	External Components	1
5	Open Problems	2
6	Revision Log	2
7	Sources	2

1. Summary

This document covers the user-interface peripherals: the display, the front navigation inputs, and the side controls. All inputs connect to ESP32-S3 GPIOs; the display connects over SPI.

Peripheral	Qty	Part / Status
Display	1	2.4 in IPS, 240 × 320, SPI, bare panel on FPC. Panel not yet selected.
Front tactile switches (up / down)	2	TS126CJ
Rotary encoder	1	481RV12172100
Side switches (power, volume up, volume down)	3	E-Switch TL1014BF220QG, side-actuated SMD

2. Design Requirements

Requirement	Value
Display	2.4 in IPS, 240 × 320, SPI; ST7789 or ILI9341 driver; glass within ≈ 43 × 60 mm (inside the battery footprint); vendor must publish the FPC pinout and backlight configuration
Front navigation	Two tactile switches (up, down) and one rotary encoder
Side controls	Power, volume up and volume down; right-angle switches on the enclosure side
Wake from standby	Any input that wakes the player must use an RTC-capable ESP32-S3 GPIO
Enclosure	≈ 56 × 75 × 13.2 mm; switch and encoder heights must fit
Constraints	Low noise · Small footprint

3. Interface

Peripheral	Connection
Tactile and side switches	One ESP32-S3 GPIO each, pulled up, switch to GND; debounced in firmware or with an RC filter
Rotary encoder A / B	Two ESP32-S3 GPIOs decoded by the pulse counter (PCNT) peripheral; pull-ups and RC filtering per the encoder datasheet
Display	SPI (SCLK, MOSI, CS, D/C), reset, and a backlight control output

4. External Components

Ref	Part	Notes	Qty
SW_UP, SW_DN	TS126CJ	Front tactile switches	2
ENC	Würth 481RV12172100	Listed as the encoder; the part is a WS-RPTL 20 × 20 mm illuminated 8-position rotary switch with a DPDT push switch (see Open Problems)	1
SW_PWR, SW_VOLU, SW_VOLD	E-Switch TL1014BF220QG	Side-actuated, right-angle SMD; 4.7 × 3.5 mm; 220 gf; 50 mA at 12 V DC. The enclosure button presses the actuator.	3
LCD	TBD	2.4 in IPS SPI panel with FPC connector	1

5. Open Problems

#	Problem
1	Display panel selection: driver IC, FPC pitch and pin count, backlight configuration (parallel LEDs need no boost converter; a series string does).
2	481RV12172100 is an 8-position rotary switch (1P8T) with a push switch and bicolour LED, not an incremental quadrature encoder, and measures 20 × 20 mm. Confirm it or replace it with a low-profile incremental encoder.
3	Confirm TS126CJ datasheet values: dimensions, height, operating force and life.
4	GPIO assignment, including RTC-capable pins for every wake source.

6. Revision Log

Rev	Date	Change
0	2026-09-27	Initial issue: peripheral set, requirements, interface, selected front switches and encoder.
1	2026-09-27	Side switches: E-Switch TL1014BF220QG. Encoder part type flagged.

7. Sources

1. [TS126CJ Product Page \(DigiKey\)](#)
2. [481RV12172100 Product Page \(DigiKey\)](#)
3. [481RV12172100 WS-RPTL Rotary Switch \(RS Components\)](#)
4. [TL1014BF220QG Product Page and Datasheet \(DigiKey\)](#)